

Chapter 6

Phonology of the Celtic languages

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This chapter offers a review of the phonological systems of the extant Celtic languages, including their consonant and vowel inventories as well as their salient phonological properties. The chapter focuses especially on how these phonologies differ from each other and on the similar developments that have occurred among them.

1 Introduction

In this chapter, I offer an overview of the synchronic phonology of the modern Celtic languages. My goal is to see what is common to these languages and to take seriously the proposition that there might be a more general Celtic phonological system that characterizes all of these languages together. The individual languages vary, of course, but I proceed on the hypothesis that the variations are in balance and that the differences between the languages aren't random but are tempered by other things.

The organization of this chapter is as follows. First, I lay out the general historical and geographic context for the modern Celtic languages. I then turn to a *very* brief overview of the phonological systems of each of the Celtic languages, focusing on points of similarity and comparison. I then try to synthesize these different systems and discuss the common threads that run through the phonologies of all the languages.

There are many many issues I simply cannot cover here given the breadth of the topic. Thus, my goal is to give enough of a sense of these individual systems to address the larger question, so that interested readers can pursue specific topics further on their own.



2 Overview and context

Celtic is a branch of the Indo-European family. The modern representatives of this group include Breton, Cornish, Irish, Manx, Scottish Gaelic, and Welsh. Breton, Cornish, and Welsh comprise the Brythonic subgroup and Irish, Manx, and Scottish Gaelic the Goidelic subgroup (Fife 2010).

Celtic languages used to be spoken across much of Europe but now are spoken only in Ireland, Britain, and France (Eska 2010).¹ It's generally thought that Celtic peoples migrated from the European continent to the British islands and their languages then died out on the continent. The Breton language in France resulted from Celtic migration *back* to the continent.

Subsequent migrations or invasions by various Germanic peoples, Romans, and Norman French forced the various Celtic groups to the peripheries of Britain, Ireland, and France. In all cases, the languages are in significant and increasing competition with large national languages.

The speaker populations are all fairly small and difficult to quantify: Do we include all speakers or restrict ourselves to first language speakers? Do we only include speakers who use the language regularly? Do we expect speakers to be literate in their language? Table 1 gives *very* rough estimates of speaker populations for each language, based on a fairly generous interpretation of native speaker.²

Table 1: Estimated speakers

Language	Speakers
Breton	200,000
Cornish	0
Irish	40,000
Manx	0
Scottish Gaelic	50,000
Welsh	500,000

The numbers for Cornish and Manx require some context. Both of them are formerly dormant languages that are currently undergoing active revitalization

¹There are, of course, expatriate communities in other places. The most significant ones are: Scottish Gaelic in Nova Scotia, Canada and Welsh in Patagonia, Argentina.

²More official, though perhaps less accurate, estimates are available at <https://www.ethnologue.com>.

efforts (Lowe 2018, Lewin 2022). The last first-language native speaker of Manx, Ned Maddrell, died in 1974. The last first-language native speaker of Cornish, Dolly Pentreath, died in 1777. Both languages have a number of second language speakers, probably numbering near 1000 in each case. In the case of Manx, there are reportedly new first-language native speakers. The first-language situation with Cornish is unknown at this time.

The Celtic languages exhibit a number of frequently cited typological features that are fairly rare (Fife 2010). These include the following:

1. consonant mutation
2. inflected prepositions
3. VSO word order

Other properties are occasionally cited as well, but these three are generally considered to be core Celtic properties. In order to get a sense of the larger grammatical setting within which each phonology is seated, each of these properties are investigated in turn.

The consonant mutations involve systematic changes to the initial consonant of a word in specific morphological or syntactic contexts. For example, in all the Celtic languages, the initial consonant of a feminine singular noun will change after the definite article.³ This consonant change is overtly encoded in the orthographies of these languages, as seen in Table 2.

Table 2: Consonant mutation with feminine nouns after the definite article

Language	Noun	Gloss	With article
Breton	<i>bro</i>	‘country’	<i>ar vro</i>
Cornish	<i>tus</i>	‘people’	<i>an dus</i>
Irish	<i>bean</i>	‘woman’	<i>an bhean</i>
Manx	<i>ben</i>	‘woman’	<i>yn ven</i>
Scottish Gaelic	<i>caora</i>	‘sheep’	<i>a’ chaora</i>
Welsh	<i>cath</i>	‘cat’	<i>y gath</i>

³The Goidelic languages distinguish grammatical case using the mutations; in those languages, the generalization applies only in the nominative.

The key fact is that words that begin with the same consonant mutate the same way in the same environment. Thus, for example, any feminine singular noun that begins with [b] in Breton mutates to [v] after the definite article. There are additional niceties in each language, which are discussed below.

All the modern Celtic languages exhibit inflected prepositions, where some prepositions are marked for the person and number of their objects. In Table 3 and Table 4 we see the relevant orthographic forms for the cognate preposition ‘on’. These may or may not co-occur with the relevant overt pronoun. For example, in Scottish Gaelic, the overt pronoun does not co-occur with an inflected preposition; in Welsh, an overt pronoun does not occur in formal registers but typically does occur in the spoken language, e.g. *arna i* ‘on me’.

Table 3: Inflected preposition ‘on’ in Breton, Cornish, and Irish

	Breton	Cornish	Irish
1SG	<i>warnon</i>	<i>warnav/m</i>	<i>orm</i>
2SG	<i>warnout</i>	<i>warnas</i>	<i>ort</i>
3SGM	<i>warnañ</i>	<i>warnadho</i>	<i>air</i>
3SGF	<i>warni</i>	<i>warnedhi</i>	<i>uirthi</i>
1PL	<i>warnomp</i>	<i>warnan</i>	<i>orainn</i>
2PL	<i>warnoc’h</i>	<i>warowgh</i>	<i>oraibh</i>
3PL	<i>warno</i>	<i>warnedhans</i>	<i>orthu</i>

Table 4: Inflected preposition ‘on’ in Manx, Scottish Gaelic, and Welsh

	Manx	SG	Welsh
1SG	<i>orrym</i>	<i>orm</i>	<i>arna</i>
2SG	<i>ort</i>	<i>ort</i>	<i>arnot</i>
3SGM	<i>er</i>	<i>air</i>	<i>arno</i>
3SGF	<i>urree</i>	<i>oirre</i>	<i>arni</i>
1PL	<i>orrin</i>	<i>oirnn</i>	<i>arnon</i>
2PL	<i>erriu</i>	<i>oirbh</i>	<i>arnoch</i>
3PL	<i>orroo</i>	<i>orra</i>	<i>arnyn</i>

The Celtic languages are also noted for VSO (verb–subject–object) word order. In all of them, this occurs when the verb is directly inflected (e.g., in the simple past tense). Following are examples in Welsh (1) and in Scottish Gaelic (2).

- (1) Gwelodd y dyn y gath
saw the man the cat
‘The man saw the cat.’
- (2) Chunnaic an duine an cat
saw the man the cat
‘The man saw the cat.’

As previously noted, Cornish and Manx are actively undergoing revitalization, and this surely has consequences for how researchers might characterize the

structures of these languages. However, it's fair to say that the other languages are also strongly affected by their contact with neighboring majority languages. To put it bluntly, natural conversations in these languages almost always involve code-switching or nonce borrowing with the neighboring dominant language.

For example, in speaking Welsh casually and talking about 'voting in the election', one is far more likely to hear borrowed or code-switched *fotio yn y lecsiwn* [vɔtjo ən ə lɛkʃɨn], rather than native *pleidleisio yn yr etholiad* [p^hlejdlejšo ən ər ɛθɔljad].

This general situation has obvious consequences for the phonologies of these languages. The phonologies don't exist autonomously but in a, perhaps sometimes conflicting, relationship with other phonologies. For example, Bennett et al. (2018) discuss how the palatalization contrast is being lost in various Irish dialects, presumably as an effect of contact with English. Morris (2021) discusses a very interesting example where some Welsh varieties have acquired English [ɹ], but some English varieties have acquired Welsh [r, ɾ].

Keeping this in mind, I now turn to a very brief overview of the specific phonologies of each of the Celtic languages, focusing on inventory, mutation, and any salient phonological processes. The systems are treated in two groups. The Brythonic or P-Celtic languages can be opposed to the Goidelic or Q-Celtic languages. As noted above, the Brythonic languages are comprised of Breton, Cornish, and Welsh; the Goidelic languages include Irish, Manx, and Scottish Gaelic. The P- and Q- nomenclature reflects the fact that Indo-European *k^w generally corresponds to Brythonic [p] and Goidelic [k] (written as either <c> or <q> in the orthography). Table 5 provides some examples of these correspondences in word-initial position.

Table 5: Reflexes of Indo-European *k^w

	'children'	'five'	'what'	'four'
P-Celtic				
Breton	<i>(bugale)</i>	<i>pemp</i>	<i>pe</i>	<i>pevar</i>
Cornish	<i>(flehes)</i>	<i>pemp/pymp</i>	<i>pe/py</i>	<i>peswar</i>
Welsh	<i>plant</i>	<i>pump</i>	<i>pa</i>	<i>pedwar</i>
Q-Celtic				
Irish	<i>clann</i>	<i>cúig</i>	<i>cén</i>	<i>ceithre</i>
Manx	<i>cloan</i>	<i>queig</i>	<i>cre</i>	<i>kiare</i>
Scottish Gaelic	<i>clann</i>	<i>còig</i>	<i>(dè)</i>	<i>ceithir</i>

3 Goidelic

The phonologies of the Goidelic languages exhibit some commonalities. First, all the languages exhibit a palatalization contrast among the consonants, an innovation from Proto-Celtic. Second, the mutation systems are somewhat simpler than in the Brythonic languages and include at most two mutations, traditionally termed *lenition* and *eclipsis*.⁴ On the other hand, the mutations affect more consonants and the mappings are somewhat more dramatic. Third, all the languages exhibit vowel reduction of stressless short vowels. Finally, the languages all include a marked epenthesis process involving clusters of a sonorant and an obstruent.⁵

3.1 Irish

Irish is spoken in Ireland, principally the Republic of Ireland, but to a lesser extent in Northern Ireland as well. The densest speaker populations are in the *Gaeltacht* or rural areas predominantly along the west coast of Ireland in the counties of Donegal, Galway, Kerry, and Mayo. In the south and east, Cork, Waterford, and Meath are official Gaeltacht areas as well. There are substantial numbers of second language speakers in other parts of the country, especially Dublin. There are also what we might term “new” speakers, speakers whose proficiency comes from immersion at school or in communities, but whose parents are not speakers. This is an important category for the other Celtic languages as well.

The language exhibits substantial dialect variation. Traditionally, three main dialect areas are distinguished.

- Munster (Cork, Kerry, Waterford)
- Connacht (Galway, Mayo)
- Ulster (Donegal)

There are interesting phonological differences between the dialects, but I will largely set these aside given the scope of the chapter.

There are excellent comprehensive phonological treatments of the language, including Ní Chiosáin (1991), Green (1997), Hickey (2014), and Ó Siadhail (1989). Generalizing across the dialects, the consonants are given in Table 6. Notice that

⁴This is the traditional analysis. Some analysts posit many more, e.g. Carnie (2008) for Irish.

⁵As I show later, the process appears to have occurred in Manx, but does not seem to be active in the synchronic language.

there is a contrast between plain (properly, velarized) and palatalized consonants for virtually all consonants. As well, in some dialects the consonants include velarized versions of some or all of [l, r, n] that, together with palatalization, makes for a four-way contrast among these segments.

Table 6: Irish consonants

	Labial	Coronal	Palatalized				Dorsal	Glottal
			Labial	Coronal	Dorsal	Dorsal		
Stop	p, b	t, d	p ^j , b ^j	t ^j , d ^j	k ^j , g ^j	k, g		
Nasal	m	n	m ^j	n ^j	ŋ ^j	ŋ		
Fric.	f, v	s	f ^j , v ^j	s ^j	x ^j , j	x, γ		h
Approx.		l, r		l ^j , r ^j				

The Irish vowels appear in Table 7. Note that there is a contrast for length at all heights with some qualitative effects, though it's unclear whether this contrast extends to the low vowel(s). Ní Chiosáin (1991) proposes a much sparser input inventory for the short vowels.

Table 7: Irish vowels

	Front	Central	Back
High	i, i:		ɯ, u:
Mid	e, e:	ə	ɔ, o:
Low		a, a:	

Historically, the palatalization contrast for the consonants can be connected to the vowels, but the vowel system has undergone such radical changes that this is no longer the case for long vowels, and the contrast is phonemic synchronically. Short non-low vowels do show a relationship: they are front or back depending on the palatalization value of the following segment. Ní Chiosáin (1991: 141) cites the examples in Table 8.

There are a number of alternations between palatalized and plain consonants, but these are all morphologized. The only non-morphologized part of these alternations is that they are transitive. When a consonant becomes palatalized or non-palatalized, adjacent consonants to the left follow suit. In other words, sequences of adjacent consonants agree in palatalization.

Table 8: Backness of a non-low vowel determined by the following consonant in Irish

<i>cur</i>	[kur]	‘put’	nonfin.
<i>cuir</i>	[kir ⁱ]	‘put’	imper.
<i>muc</i>	[muk]	‘a pig’	nom. sg.
<i>mhuic</i>	[wik ⁱ]	‘a pig’	dat. sg.

Like all the other Celtic languages, Irish has initial consonant mutation. As previously noted, the initial consonant of a word undergoes systematic changes depending on its morphosyntactic environment. Setting aside the nature of the environment, traditionally there are two basic mutation patterns in Irish.⁶ Following traditional terminology, these are referred to as *lenition* and *eclipsis*. Both are laid out in Table 9. Basically, lenition converts stops to fricatives; compare examples (3–4) and (5–6). In addition, voiceless fricatives either delete or debuccalize. Coronal consonants also debuccalize. Finally, [m] and [mⁱ] become fricatives as well. Eclipsis makes voiceless stops voiced and makes voiced stops into nasals. In addition, the fricatives [f] and [fⁱ] become voiced. These changes are overtly expressed in the orthography as well.⁷

As already noted, the mutations are triggered by morphosyntactic environments so they are curiously morphosyntactic and phonological at the same time. On the one hand, the morphosyntactic conditioning suggests that they should be viewed as something morphological. On the other hand, the fact that the consequences of mutation are phonological suggests that it should be viewed as a phonological process.

There is a virtual industry in debating which of these is correct for different mutations in the different Celtic languages from myriad theoretical perspectives. See, for example: Hamp (1951), Rogers (1972), Anderson (1976), Lieber (1992), Ball & Müller (1992), Kibre (1997), Pyatt (1997), Stewart (2004), Green (2006), Hannahs (2011a, 2013). I will not address this question here and instead focus on the phonological *consequences* of this process.

The most obvious consequence of these is that they alter the set of possible word-initial consonants and clusters (Grijzenhout 1995, Ní Chiosáin 1999). Thus, for example, [x] is not generally allowed word-initially in content words but is permitted word-initially in lenition contexts, as shown in the past tense forms (4) and (6).

⁶Though see Carnie (2008) for a more complex picture.

⁷Note that palatalization is expressed with vowel symbols in the spelling.

Table 9: Irish consonant mutation

Phonology			Orthography		
Base	Lenition	Eclipsis	Base	Lenition	Eclipsis
p	f	b	p	ph	bp
t	h	d	t	th	dt
t ^j	h ^j /x ^j	d ^j	t	th	dt
k	x	g	c	ch	gc
k ^j	x ^j	g ^j	c	ch	gc
b	v	m	b	bh	mb
b ^j	v ^j	m ^j	b	bh	mb
d	ɣ	n	d	dh	nd
d ^j	j	n ^j	d	dh	nd
g	ɣ	ŋ	g	gh	ng
g ^j	j	ŋ ^j	g	gh	ng
f	Ø	v	f	fh	bhf
f ^j	Ø	v ^j	f	fh	bhf
s	h	s	s	sh	s
s ^j	h ^j /h/x ^j	s ^j	s	sh	s
m	v	m	m	mh	m
m ^j	v ^j	m	m	mh	m

- (3) caithim é
throw-1sg it
[kaim e:]
'I throw it.'
- (4) chaith mé é
threw I it
[xa(j) me: e:]
'I threw it.'
- (5) caillim é
lose-1sg it
[kal^jim e:]
'I lose it.'

- (6) chaill mé é
lost I it
[xal^h me: e:]
'I lost it.'

Similarly, the mutations create word-initial clusters not otherwise allowed outside of mutation contexts. For example, while [sn] and [s^hl^h] are generally legal onsets, [hn] and [h^hl^h] are only legal in lenition contexts; they are not otherwise acceptable. Compare (7) and (9) with (8) and (10); again, all in the past tense:

- (7) snámhaim
swim-1sg
[sna:wim]
'I swim.'
- (8) shnámh mé
swam I
[hnɔv me:]
'I swam.'
- (9) sleamhnaím
slip-1sg
[s^hl^hawni:m]
'I slip.'
- (10) shleamhnaigh mé
slipped I
[h^hl^hawnə me:]
'I slipped.'

These sorts of effects on the inventory of (word-initial) segments and the inventory of word-initial consonant clusters run through all the Celtic languages.

Another very salient aspect of Irish phonology that has received a lot of attention in the literature is epenthesis into rising sonority clusters.⁸ The phenomenon is curious in a number of respects. First, the clusters that this applies to are composed of a sonorant followed by a voiced consonant or a voiceless fricative; epenthesis is blocked when the second consonant is a voiceless stop or if the consonants are homorganic. The context following the cluster is irrelevant. In Table 10, I give examples where epenthesis occurs (from Ní Chiosáin 1999: 560).

⁸In the literature, this is sometimes referred to as *svarabhakti*.

In Table 11, I give examples where epenthesis does not occur because the second consonant is a voiceless stop (from Ní Chiosáin 1999: 561). And in Table 12, I give cases where epenthesis is blocked because the two consonants are homorganic.

Table 10: Examples of Irish epenthesis

lb	<i>bolb</i>	[boləb]	‘caterpillar’
lw	<i>gealbhan</i>	[gʲaləwən]	‘sparrow’
lx	<i>tulchach</i>	[tuləxəx]	‘hilly’
rb	<i>borb</i>	[borəb]	‘abrupt’
rm	<i>gorm</i>	[gorəm]	‘blue’
rx	<i>dorcha</i>	[dorəxə]	‘dark’
nb	<i>binb</i>	[bʲinʲibʲ]	‘venom’
nm	<i>ainm</i>	[anʲimʲ]	‘name’
nv	<i>leanbh</i>	[lʲanəv]	‘child’

Table 11: Voiceless stop examples (no Irish epenthesis)

rp	<i>corp</i>	[korp]	‘body’
rt	<i>gort</i>	[gort]	‘field’
rk	<i>cearc</i>	[kʲark]	‘hen’
lp	<i>spalp</i>	[spalp]	‘burst forth’
lt	<i>alt</i>	[alt]	‘joint’
lk	<i>olc</i>	[olk]	‘evil’
nt	<i>cantal</i>	[kantəl]	‘irritation’

Table 12: Homorganic consonant examples (no Irish epenthesis)

lr	<i>iolra</i>	[olrə]	‘plural’
ld	<i>gallda</i>	[galda]	‘foreign’
ls	<i>foilsigh</i>	[folʲsʲi]	‘reveal’
mb	<i>siombalach</i>	[sʲombələx]	‘symbolic’
nd	<i>gandal</i>	[gandal]	‘gander’
ng	<i>teanga</i>	[tʲaŋgə]	‘language’

Finally, the process also cannot occur when the preceding vowel is long. Ní Chiosáin (1991) cites these examples: *téarma* [tʲe:rmə] ‘term’ and *léargas* [lʲe:rgəs] ‘insight’.

This is a devilishly tricky phenomenon to treat. First, the process exhibits no alternations. The first consonant must be a coda, but epenthesis doesn’t care whether the second consonant is tautosyllabic or heterosyllabic. There are no processes that bear on the consonantal restrictions. Finally, the length of the preceding vowel doesn’t alternate. The main evidence for the process is the surface absence of clusters where epenthesis must apply (but does not).

An additional argument can be made from the morphology (Ní Chiosáin 1999). When a final consonant is palatalized for morphological reasons, this palatalization also occurs on the consonant preceding the epenthetic vowel, as if the consonants were adjacent when palatalization occurs, e.g. in *colm* ‘dove’ [koləm], but *coilm* ‘doves’ [kelʲimʲ].

A second complication is that syllable structure doesn’t matter, as shown by the irrelevance of what follows the cluster. Yet moraic status *does* matter as far as the preceding vowel (since it cannot be long). Finally, the odd restriction on voiceless stops presents an additional complication (Fullwood 2013). I return to the process of epenthesis further on because very similar facts obtain in Scottish Gaelic.

Stress in most dialects of Irish falls on the initial syllable, except for a few borrowings. However, stress in Munster Irish is quite striking and has received a fair amount of attention in the theoretical literature, e.g. Iosad (2013) and Bennett (2017).

In Munster, stress falls on the first syllable, e.g. in *clagarnach* [ˈkʲlʲagərnəx] ‘clattering’. If one of the first three syllables contains a long vowel or diphthong, then it gets stress, e.g. in *cailín* [kaˈlʲi:nʲ] ‘girl’, *amadántaíocht* [əməˈdɑ:nti:xt] ‘foolishness’.

There are two interesting additional complications. First, if the first two syllables are long, then stress falls on the second one, e.g. *díomhaoin* [dʲiːˈvi:nʲ] ‘idle’. Otherwise stress falls on the first long vowel in the first three syllables, e.g. *am-paránaíocht* [ˈoʊmpərəːni:xt] ‘ungainliness’.

Second, if there are no long vowels or diphthongs in the first three syllables and the second vowel is [a] followed by [x], then it gets stressed, e.g. in *bacach* [bəˈkax] ‘beggar’, *bacacha* [bəˈkaxə] ‘beggars’, *casachtach* [kəˈsaxtəx] ‘coughing’.

Finally, a comment on syllable structure. It’s been observed that a single intervocalic consonant affiliates to the syllable to the left in Irish, rather than to the right. Ní Chiosáin et al. (2012) investigate this claim experimentally and find that, in fact, affiliation to the left is preferred when the preceding vowel is short, and

that the consonant typically affiliates in *both* directions. Strikingly, very similar facts have been cited for English (Hammond 1997; 1999). Notice that the stress facts just cited for Munster [ax] sequences lend support to such an account, as [ax] would constitute a syllable rhyme on this view.

3.2 Manx

The Manx language is another Goidelic language very closely related to Irish. As already noted above, Manx is spoken on the Isle of Man and went dormant upon the death of the last native speaker (at that time) in 1974. However, also as previously noted, Manx is currently the object of active revitalization efforts and there are now over a thousand speakers, including children.

As with Irish, I outline the segmental inventory, the mutations, and salient aspects of the phonology. Where appropriate, I compare these with the Irish system above. It's important to keep in mind, though, that Manx is in great flux due to the recent revitalization efforts. My description is based mostly on Broderick (1984).

The Manx consonants appear in Table 13. The system is quite similar to that of Irish, though note the absence of palatalized labials. Note also that the coronal stops are dental and occasionally affricated.

Table 13: Manx consonants

	Labial	Coronal	Palatalized		Dorsal	Glottal
			Coronal	Dorsal		
Stop	p, b	t̪, d̪	tʲ, dʲ	kʲ, gʲ	k, g	
Fricative	f, v	s	sʲ	xʲ	x	h
Nasal	m	n	nʲ	ɲʲ	ɲ	
Approximant	w	l, r	lʲ, j			

Manx has a simple five-vowel system with a length contrast and schwa. The vowels appear in Table 14.

There is a sporadic process of intervocalic weakening of stops, but it's not clear under what conditions this occurs and what the precise targets are. For example, a word like *scapail* 'escaping' can be pronounced [skəp^he:lʲ], [skəβe:lʲ], or [skəve:lʲ]. Similarly, *brattag* 'flag' can be pronounced [bratag], [bradag], or [braðag]. Voiced stops exhibit a similar pattern. For example: *chibbyrt* 'well' as [tʲubərt], [tʲuβərt], or [tʲuvərt]. The lenition mutation occurred historically

Table 14: Manx vowels

	Front	Central	Back
High	i, i:		u, u:
Mid	e, e:	ə	o, o:
Low		a, a:	

within words and it's reasonable to suppose this weakening process is a similar reassertion of that pattern.

Another salient process is prestopping where nasals and the liquid [l] can acquire a preceding voiced stop when the preceding vowel is stressed, as exemplified in Table 15. Interestingly, as we'll see later, a similar process occurs in Cornish.

Table 15: Manx prestopping

<i>cam</i>	'crooked'	[kam]	[kabm]
<i>eeym</i>	'butter'		[i:bm]
<i>bane</i>	'white'	[be:n]	[be(:)dn]
<i>keayn</i>	'sea'		[kidn]
<i>lhong</i>	'ship'	[loŋ]	[lo/ugŋ]
<i>shooyil</i>	'walking'	[s ^h u:l]	[s ^h u:dl]
<i>keeill</i>	'church'	[k ^h i:l ^h]	[k ^h idl ^h]

Unlike the other Goidelic languages, Manx does not seem to exhibit epenthesis. There are many examples of forms that would undergo epenthesis in the other Goidelic languages that do not do so in Manx. See Table 16 for examples where no epenthesis appears in Manx but does appear in the corresponding Irish form. On the other hand, there are a number of cases where the cognate form in Manx has apparently undergone epenthesis (see Table 17). It would seem reasonable to suppose that Manx once had epenthesis, but it has since been lost leaving now lexicalized remnants.

Manx mutations are given in Table 18. It's important to keep in mind that eclipsis is virtually absent from the modern language.⁹

⁹Eclipsis occurs only after the article in the genitive plural and after the plural possessive pronoun *nyn*.

Table 16: No epenthesis in Manx

Manx			Irish	
<i>jiarg</i>	[dʲarg]	‘red’	<i>dearg</i>	[dʲarəɟ]
<i>pershagh</i>	[pɛrʒəx]	‘peach’		
<i>argid</i>	[ərgəd]	‘money’	<i>airgead</i>	[æɾʲiɟʲəd]
<i>lorg</i>	[lɔrg]	‘pole’		
<i>margher</i>	[margɛ]	‘market’		

Table 17: Lexicalized epenthesis in Manx

Manx			Irish	
<i>gorrym</i>	[gɔrəm]	‘blue’	<i>gorm</i>	[gɔrəm]
<i>dorraghey</i>	[dɔrəxə]	‘dark’	<i>dorcha</i>	[dɔrəxə]
<i>ennym</i>	[ɛnəm]	‘name’	<i>ainm</i>	[æɾʲim]
<i>(thie) fillym</i>	[filəm]	‘cinema’		

3.3 Scottish Gaelic

Scottish Gaelic is the third Goidelic language.¹⁰ The language formerly extended over much of Scotland, but is now fairly restricted to the extreme northern and northwestern peripheries and islands. There was substantial emigration during the highland clearances in the 18th and 19th centuries, but, except perhaps for Nova Scotia and Prince Edward Island in Canada, there are no current substantial speaker communities outside of Scotland.

The language is the object of active revitalization efforts and there are now Gaelic medium schools in Edinburgh and Glasgow, as well as in the north. Historically, there is a great deal of dialect variation, but this is greatly reduced as a function of language attrition and media in English. There are a number of useful discussions of the phonology and phonetics of Scottish Gaelic in the literature and my discussion draws on a number of these, for example: Bauer (2011), Bosch (1998), Bosch (1991), Bosch & de Jong (1997), Clements (1986), Green (1997), Ladefoged et al. (1998), Hammond et al. (2014), Hammond et al. (2017), Nance (2015), Smith (1999).

¹⁰The language is occasionally referred to as “Scots Gaelic” but this term is dispreferred by speakers of the language.

Table 18: Manx mutations

Base	Lenition	Eclipsis
p	f	b
t	h	d
t ^j	h/x ^j	d ^j
k	x	g
k ^j	x ^j	g ^j
b	v	m
bw	w	mw
m	v	
mw	w	
d	ɣ	n
d ^j	j	n ^j
g	ɣ	ŋ
g ^j	j	ŋ ^j
f	∅	v
fw	hw	w
s	h	
s ^j	h/x ^j	

The Scottish Gaelic consonants appear in Table 19. Note a palatalization contrast as in Irish and Manx. Note, too, the presence of a velarized series for sonorant coronals. These are not all present in all dialects and the contrast is occasionally realized instead in terms of length and/or diphthongization of the preceding vowel. The voicing contrast for stops is different from the other Goidelic languages. The contrast in initial position is between aspirated and unaspirated voiceless stops. Medially, the unaspirated series is typically voiced, and the aspirated series is typically *preaspirated* in non-initial position, e.g. in *cat* [k^ha^ht] ‘cat’.¹¹

The Scottish Gaelic vowels appear in Table 20. Note the presence of length and nasality contrasts, though the nasality contrast is not extensive and may be being lost among younger speakers. Note, too, that there are non-low back unrounded vowels in the system.¹²

¹¹The word-initial version of this second series is what appears in Table 19.

¹²Chasaide (1995) reports that [ɣ] occurs in some varieties of Irish.

Table 19: Scottish Gaelic consonants

	Labial	Dental	Alveolar	Palatalized		Dorsal	Glottal
				Coronal	Dorsal		
Stop	p ^h , p	t ^h , t		t ^{jh} , t ^j	k ^{jh} , k ^j	k ^h , k	
Nasal	m		n	ɲ		n ^y	
Fricative	f, v		s	ʃ	ç, ʝ	x, ɣ	h
Lateral			l	l ^j		l ^y	
Rhotic			r	r ^j		r ^y	
Approx				j			

Table 20: Scottish Gaelic vowels

	Front	Central	Back	
			Round	Unround
High	i, i:, ĭ, ĭ:		u, u:, ũ, ũ:	u:, u:, ũ:, ũ:
Mid High	e, e:		o, o:	ɤ, ɤ:
Mid Low	ɛ, ɛ:, ě, ě:		ɔ, ɔ:, ɔ̃, ɔ̃:	
High		a, a:, ă, ă:		

The Scottish Gaelic lenition mutation is given in Table 21. The effects are essentially the same as in Irish and Manx. Note that in dialects with the velarization contrast, lenition may also affect these consonants such that sonorant coronals lose velarization and palatalization in the relevant contexts. When this occurs, it is not represented in the orthography.

Eclipsis mutation is absent in many dialects and not reflected in the orthography. If it does occur, it has the effects in Table 22. The expected pattern is shown in the Eclipsis column. However, Ó Maolalaigh (1995) notes additional patterns, which appear in different dialects; these are shown in the right three columns.¹³

Another interesting difference between Scottish Gaelic and the other Goidelic languages is that the trigger for eclipsis always ends in a nasal consonant. This

¹³Ó Maolalaigh describes the patterns in terms of a voicing contrast; I've translated it into the aspiration contrast used here. It's a little unclear what Ó Maolalaigh intends by the voiced aspirated symbols. The key point about the B group is that a voicing distinction of some sort is maintained under eclipsis.

Table 21: Scottish Gaelic lenition mutation

Orthographic		Plain		Palatalized	
Unlenited	Lenited	Unlenited	Lenited	Unlenited	Lenited
p	ph	p ^h	f	—	—
b	bh	p	v	—	—
t	th	t ^h	h	t ^h	ç
d	dh	t	ɣ	t ^j	j
c	ch	k ^h	x	k ^h	ç
g	gh	k	ɣ	k ^j	j
f	fh	f	∅	—	—
s	sh	s	h	ʃ	ç
m	mh	m	v	—	—

Table 22: Scottish Gaelic mutation

Consonant	Eclipsis	A	B	C
p ^h	p	p	b ^h	m ^h
t ^h	t	t	d ^h	n ^h
k ^h	k	k	g ^h	ŋ ^h
p	m	p	p	m
t	n	t	t	n
k	ŋ	k	k	ŋ
f	v	f	f	

may be evidence that eclipsis is in the process of becoming phonological again in the language.

Finally, Scottish Gaelic exhibits essentially the same epenthesis process as Irish. See Table 23 for a few examples. Note that the inserted vowel (**bolded**) is a copy of the preceding vowel in Scottish Gaelic. In addition, there are interesting prosodic properties associated with the inserted vowel in Scottish Gaelic (Bosch & de Jong 1997, Bosch 1998). For example, Bosch & de Jong (1997) report that the epenthetic vowel in Barra Scottish Gaelic is acoustically stressed.

The process is blocked in the same contexts as in Irish as shown in Table 24: when the first consonant is not a sonorant, when the preceding vowel is long or a

Table 23: Epenthesis (bolded) in Scottish Gaelic

<i>seanmhair</i>	[ʃɛnɛ v ər]	‘grandmother’
<i>arm</i>	[a r am]	‘army’
<i>dorcha</i>	[dɔrɔ x ə]	‘dark’
<i>tilg</i>	[tʲɪ l ɪk]	‘to toss away’
<i>urchair</i>	[ʊrɔ x ɔr]	‘shot’

diphthong, when the consonants are homorganic, or when the second consonant is a voiceless stop.

Table 24: Scottish Gaelic examples where epenthesis is blocked (marked with underline)

Nonsonorant C1	<i>smachd</i>	[sma x _k]	‘stifle’
	<i>sgriobhte</i>	[skriv_ɪ f ə]	‘written’
Long vowel	<i>dualchas</i>	[duəl_ x əs]	‘tradition’
	<i>iarmailt</i>	[i:rmɛl_ɪ f]	‘firmament’
Homorganic CC	<i>Òlaind</i>	[o:lan_ d ɜ]	‘Holland’
	<i>mandrag</i>	[man_ dr ak]	‘mandrake’
C2 Voiceless Stop	<i>olc</i>	[ɔ l _k]	‘evil’
	<i>corp</i>	[kɔ r _p]	‘body’

Note how the restriction against the second consonant being a voiceless stop makes a bit more sense in Scottish Gaelic than in Irish. In Scottish Gaelic, medial voiceless stops are preaspirated. When there is a preceding sonorant, that sonorant is thus voiceless. We may then subsume the voiceless stop and homorganicity restriction under the same rubric of shared features blocking an intervening inserted element.

4 Brythonic

The Brythonic languages (Breton, Cornish, and Welsh) differ, broadly speaking, from the Goidelic languages in several respects.

1. None of the Brythonic languages make a distinction between broad and slender contrasts (a palatalization contrast). Concomitantly perhaps, none

of the Brythonic languages distinguish overt grammatical case, since this distinction is one of the markers.

2. The mutation systems in the Brythonic languages are richer and typically more direct in the phonological relationships they exhibit.
3. Except perhaps for late Cornish, none of the Brythonic languages have the kind of extensive vowel reduction evident in the Goidelic languages. Stress in Goidelic is either on the first syllable or may, in some cases and dialects, shift to a heavy syllable to the right. In the Brythonic languages, stress is typically penultimate.
4. The Brythonic languages do not exhibit the vowel epenthesis pattern we saw in Irish and Scottish Gaelic (though a different epenthesis pattern is evident in Welsh).

4.1 Breton

Breton is spoken in Brittany, France. It is unique among the Celtic languages in that it is in contact with French rather than English. However, the situation could be viewed as somewhat more complex. First, Breton speakers are thought to have migrated from Cornwall (and elsewhere) in the Middle Ages and thus, to that point, the language was under the influence of English (and its forebearers) and Latin. Second, another factor that sets it apart is that in its present setting in France, it is in some contact with the Gallo language, a Romance language closely related to French and spoken in upper Brittany.¹⁴

Breton exhibits considerable dialect variation and my characterization here attempts to generalize beyond these distinctions. Fuller descriptions of the general system can be found in Hemon (1995) and Press (2011).

The Breton consonant inventory appears in Table 25. Note the absence of a plain versus palatalized contrast and the presence of labio-velar stops.

It's worth noting that the rhotics [r, ɾ, ʀ] are extremely variable with some speakers producing taps and other assimilating fully to a French uvular. An unresolved issue is whether some obstruents contrast in fortis-lenis and/or length in at least some dialects.¹⁵

¹⁴Interestingly, Cornish shows evidence of French influence as well, at the very least because of the Norman conquest, but also because of continuing contact with Breton (Ferdinand 2013). Thus, one would not want to say that French influence on Breton began with its transposition to France.

¹⁵See Press (2011) for discussion.

Table 25: Breton consonants

	Labial	Coronal	Pal-Alv	Dorsal	Lab-Dors	Uvular/Glottal
Stop	p, b	t, d		k, g	k ^w , g ^w	
Fricative	f, v	s, z	ʃ, ʒ	x, ɣ		h
Nasal	m	n	ɲ			
Approx		l, r, ɹ	ʎ, j, ɥ		w	ʁ

A very salient consonant process in the language is final devoicing and voicing assimilation. Obstruents are voiceless in absolute phrase-final position but assimilate in voicing to a following segment. Compare (11) with (12–15) from Hemon (1995: 79).

- (11) sellit ouzh ar beleg [sɛlid uz aʁ bɛ:lɛk]
 look at the priest
 ‘Look at the priest.’
- (12) setu amañ ar beleg kozh [sety amã aʁ bɛ:lɛk ko:s]
 voila here the priest old
 ‘Here is the old priest.’
- (13) beleg ar barrez [bɛ:lɛg aʁ baʁɛs]
 priest the parish
 ‘the priest of the parish’
- (14) ur beleg mat [œʁ bɛ:lɛg ma:t]
 a priest good
 ‘a good priest’
- (15) ur beleg mat eo [œʁ bɛ:lɛg ma:d eo]
 a priest good is
 ‘He is a good priest.’

Strikingly, if two identical obstruents come together, they surface as voiceless (as shown in 16 and 17; again, from Hemon 1995). This happens whether they initially agree in voicing or not.

- (16) bloas zo [blwa(s) so]
 year ago
 ‘a year ago’

- (17) dek gwele [de:k kwe:le]
 10 beds
 ‘10 beds’

However, the outcome does vary by dialect when both consonants are initially voiceless (Le Dû 1986). Interestingly, this same restriction is also part of Welsh classical meter, though it is *not* a part of the synchronic language (Morris-Jones 1980, ap Dafydd 1994, Llwyd 2007).

The Breton vowel inventory appears in Table 26. Note the presence of front round vowels and a contrast between open and close among mid vowels. Note too the robust nasalization contrast.

Table 26: Breton vowels

	Front		Back	
	Unround	Round	Unround	Round
High	i, ĭ	y, ỹ		u, ũ
Close-mid	e, ě	ø, ȯ		o, ȯ
Open-mid	ɛ, ẽ	œ, œ̃		ɔ, ɔ̃
Low	a, ă		ɑ, ă	

The Breton mutations appear in Table 27. The pattern is similar across the Brythonic languages: more mutations and a more transparent phonological relationship between each sound and what it mutates to.

There are a number of interesting restrictions on the Breton mutation system. Some are reminiscent of those seen in the Goidelic systems, but some are unique to Breton. For example, *va* [ɛm] ‘my’ generally triggers the aspirate mutation; compare (18–19) and (20–21). However, when it contracts to *m* with a previous vowel-final particle, it only triggers that mutation on non-labials; compare (22) with (23).

- (18) kambr
 room
 [kambʁ]
 ‘room’

Table 27: Breton mutations

C	Mutation type			
	Soft	Aspirate	Hard	Mixed
p	b	f		
t	d	z		
k	g	x		
b	v		p	v
d	z		t	t
g	ɣ		k	ɣ
gw	w		kw	w
m	v			v

- (19) em c'hambr
to-my room-asp
[ɛm xambɾ]
'to my room'

- (20) ti
house
[ti]
'house'

- (21) em zi
to-my house-asp
[ɛm zi]
'to my house'

- (22) paner
basket
[panɛr]
'basket'

- (23) 'm paner
my basket
[m panɛr]
'my basket'

This is similar to the blocking of lenition that occurs with coronals in Irish and Scottish Gaelic (Goidelic languages). Table 28 gives examples of this lenition

process following a definite article that precedes a feminine singular noun. If the noun begins with a coronal stop, it does not mutate.¹⁶

Table 28: Lenition mutation in Irish and Scottish Gaelic. Bolded examples are cases where lenition is blocked before coronals.

Irish	<i>an bhean</i>	[a va:n]	‘the woman’
	<i>an ghal</i>	[a ɣal]	‘the vapor’
	<i>an deoch</i>	[a dʲɔx]	‘the drink’
Scottish Gaelic	<i>a’ bhean</i>	[ə vɛn]	‘the woman’
	<i>a’ chaileag</i>	[ə χalak]	‘the girl’
	<i>an dànachd</i>	[ən da:naxk]	‘the poetry’

4.2 Cornish

As already noted, Cornish became dormant upon the death of the last (at that time) native speaker in 1777. Nonetheless, it has been the subject of almost constant revitalization attempts since then. Interestingly, several different versions of modern Cornish exist, based either on middle or late Cornish. I view these simply as dialects, or better yet *chronolects*, of the same language. My main sources on the language are Jenner (2010) and Brown (2001).

The Cornish consonant inventory appears in Table 29. Note the absence of a plain versus palatalized contrast just as in Breton.

Table 29: Cornish consonants

	Labial	Dental	Alveolar	Pal-Alv	Dorsal	Glottal
Stop/Affricate	p,b	t,d		tʃ,dʒ	k,g	
Fricative	f,v	θ,ð	s,z	ʃ,ʒ	x	
Nasal	m	n			ŋ	
Approx	w,ɰ		l,r/ɹ	j		h

Late Cornish undergoes an interesting pre-stopping process that is reminiscent of that seen in Manx, except that it is reflected in the orthography. Pre-stopping in Cornish occurs after a stressed vowel and before a nasal consonant,

¹⁶It is unclear whether this occurs in Manx (Teare & Wheeler 2017).

e.g. *kabm* [kabm] ‘step’ and *pedn* [pɛdn] ‘head’. If the nasal occurs in a stressless syllable, pre-stopping does not occur, e.g. *perghen* [pɛrɛn] ‘owner’ and *pedrevan* [pɛˈdrɛvən] ‘lizard, newt’.

Since stress shifts under suffixation, this creates alternations in the plural, e.g. in *kabmas* [ˈkabmas] ‘bay, bend’ vs. *kamajow* [kaˈmadʒow] ‘bays, bends’ and *pednyn* [pɛdnɪn] ‘tadpole’ vs. *penydnow* [pɛˈnɪdnɔw] ‘tadpoles’.

There is evidence that the language used to undergo voicing assimilation and final devoicing as in Breton, but it’s unclear whether this occurs in the synchronic language.

The language has a number of sonority-reversed consonant clusters. These undergo epenthesis in final position, which is transparent in the Cornish orthography, but alternations occur in suffixed forms (for example, in the plural; see Table 30).¹⁷

Table 30: Epenthesis alternations in singular vs. plural forms in Cornish

vr	<i>lever</i>	[lɛvɛr]	‘book’
	<i>levrow</i>	[lɛvrow]	‘books’
tr	<i>ownter</i>	[ɔʊntɛr]	‘uncle’
	<i>owntres</i>	[ɔʊntɛrəs]	‘uncles’
dhl	<i>kenedhel</i>	[kɛnəðɛl]	‘nation’
	<i>kenedhlow</i>	[kɛnəðlow]	‘nations’
ml	<i>amal</i>	[æmɛl]	‘border, side’
	<i>emlow</i>	[ɛmlow]	‘borders, sides’

Orthographically, the epenthetic vowel is a copy of the preceding vowel, but this does not seem to be the case in the spoken form. That this is truly epenthesis, rather than deletion, is evidenced by forms that don’t alternate; e.g. *lether* [lɛθɛr] ‘letter’ vs. *letherow* [lɛθɛrow] ‘letters’.

The Cornish vowel inventory appears in Table 31. Note the length contrast and the presence of front rounded vowels. The latter only occur in middle Cornish and the corresponding chronolect.

Finally, the Cornish mutations appear in Table 32. Note that they are quite similar to the mutations in Breton.

¹⁷Bennett (2012) argues for a similar process in Irish. See Smith (1999) for Scottish Gaelic.

Table 31: Cornish vowels

	Front		Central	Back
	Unround	Round		
High	i:, ɪ	(y:, ʏ)		u:, ʊ
Mid	e:, ɛ	(ø:, œ)	ə	o:, ɔ/ʌ
Low	æ:, æ/a			ɑ:, ɒ

Table 32: Cornish mutations

C	Mutation Type			
	Soft	Aspirate	Hard	Mixed
p	b	f		
t	d	θ		
k	g	h		
b	v		p	f
d	ð		t	t
g	Ø,w		k	h
gw	w		kw	hw
m	v			f
tʃ	dʒ			

4.3 Welsh

Unsurprisingly, Welsh is spoken primarily in Wales, but there are also Welsh-speaking communities abroad, especially in England. Notably, there is a Welsh-speaking community in Argentina, where the Welsh established a colony to escape the effects of contact with the English (Bell 2018). Like the other Celtic languages, there are a number of dialects (Jones 1998, Thomas 2000, Awbery 1986), but there is a significant divide between dialects in the north and south, which have interesting phonological differences.

There is a fairly significant literature on Welsh grammar and phonology. This ranges from traditional grammars like Morris-Jones (1913) and King (2003) to more modern treatments like Hannahs (2013). There are more specific discussions of the mutation system that include Awbery (1973) and Hannahs (2011a) as well as a wonderful set of volumes that includes papers on a number of aspects

of the phonology, phonetics, and mutation system: Ball & Williams (2001), Ball & Müller (1992), and Ball & Jones (1984). Additional papers of note on other aspects of the sound system of Welsh include Hannahs (1996, 2009, 2011b), Hammond (2019), Iosad (2017b), Jones (1984), and Awbery (1984), among others.

The surface consonant inventory for Welsh is given in Table 33. Again, note the absence of a palatalization contrast. Unlike Breton and Cornish, there are no labio-velars, but it's not clear that this is a substantive difference. Cognate elements exist in Welsh, but are typically analyzed as a sequence of a velar followed by [w], e.g. *cwyno* [k^hwino] 'complain', *gwin* [gwin] 'wine', and *chwarel* [χwarel] 'quarry'.¹⁸ Some researchers recognize a series of voiceless nasals as well, but I do not represent those here (Hammond 2019). Finally, note that the back unrounded glide [ɰ] only occurs in northern dialects.

Table 33: Welsh consonants

	Labial	Dental	Alveolar	Pal-Alv	Dorsal	Uvular/Glottal
Stop/Affric	p ^h , b	t ^h , d		tʃ ^h , dʒ	k ^h , g	
Fricative	f, v	θ, ð	s, ʃ	ʃ		χ
Nasal	m	n			ŋ	
Approx	w		l, r, ɾ	j	ɰ	h

I note two interesting phonological processes. First, as in Cornish, there are final sonority-reversed clusters that are resolved by epenthesis. Unlike in Cornish, however, there are two additional restrictions. First, per Hannahs (2009, 2011b), epenthesis is at least partially conditioned by stress. Sonority-reversed clusters in words of one syllable undergo epenthesis, e.g. *llyfr* ['lɪvɪr] 'book' and *pobl* [p^hɔbɔl] 'people'. However, in longer words where the cluster appears in a stressless syllable, deletion of the final consonant is a more typical strategy, e.g. *ffenestr* ['fɛnɛst] 'window' and *posibl* [p^hɔsɪb] 'possible'. Furthermore, the final clusters [dl] and [vn] do not seem to undergo either epenthesis or deletion, e.g. *cededl* [k^hɛnɛdl] 'nation' and *ofn* [ɔvn] 'fear', as noted in Hannahs (2013) and Iosad (2017b).

It's interesting to compare this pattern with Cornish. There, the cognate for Welsh *cededl* does undergo epenthesis: *cededhel*. However, the cognate form for Welsh *ofn* is Cornish *own* [ɔvn], where the consonant (pronounced in Welsh as

¹⁸Evidence against this comes from distributional observations. For example, in unmutated onset position, [χ] is almost always followed by [w]. The only exception is *chi* [χɪ] 'you', historically *chwi* [χwi]. Similarly, while velars can be followed by [w], other segment classes cannot be.

[v]) is lost. The consonant [v] is also lost in Welsh, but at the ends of words, as in *haf* [ha(v)] ‘summer’, *cyntaf* [k^hənt^ha(v)] ‘first’, etc. Thus, the absence of epenthesis in Welsh forms like *ofn* may be due to the precarious status of [v].

Another prominent process in Welsh involves the deletion of a medial [h] when it precedes a stressless vowel (Hammond 2019). This process also affects [ɾ], with its voiced counterpart [r] occurring where [h] would be deleted. Since, as in Cornish, stress moves under suffixation, this again gives rise to alternations, e.g. *dihareb* [di^hareb] ‘proverb’ vs. *diarhebion* [dia^hɾɛbjɔn] ‘proverbs’, *eang* [‘eaŋ] ‘wide’ vs. *ehangach* [e^hhaŋaχ] ‘wider’. The *dihareb* form is particularly nice as it shows both the h/∅ and the r/ɾ alternations.

Historically, medial nasal–voiceless stop sequences were replaced with nasal–[h] sequences; these exhibit alternations as well: *cenedd* [k^hɛnɛdl] ‘nation’ vs. *cenhedloedd* [k^hɛn^hɛdlɔ(ɥ)ð] ‘nations’, *angen* [aŋɛn] ‘need’ vs. *anghenion* [aŋ^hɛnɛnɔn] ‘needs’, *tymor* [t^hɛmɔr] ‘season’ vs. *tymhorau* [t^hɛm^hɔra(ɥ)] ‘seasons’.

The surface inventory of Welsh vowels is given in Table 34. Note that the high central unrounded vowels only occur in the north.

Table 34: Welsh vowels

	Front	Central	Back
High	i:, ɪ	ɨ:, ɨ	u:, ʊ
Mid	e:, ɛ	ə	o:, ɔ
Low		a:, a	

Vowel length is only contrastive in monosyllables and then only before certain consonants. Historically, this only included [l, r, n], but this has been extended in recent borrowings where long vowels may now occur where previously only short ones did, e.g. *strôc* [stro:k] ‘stroke, shock’, *côt* [k^ho:t] ‘coat’, *grŵp* [gru:p] ‘group’, *clôs* [k^hlo:s] ‘close’, etc.¹⁹

The Welsh mutations appear in Table 35. Note that the system is not as complex as in Breton and Cornish, but still more complex than in any of the Goidelic languages. As in the other Brythonic languages, the relationships are fairly transparent phonologically. Interestingly, similar to eclipsis in the Scottish Gaelic dialects cited above, the nasal mutation can be treated phonologically, triggered only by elements that end in an overt nasal (Hammond 2019).

¹⁹The distribution of vowel length in Welsh is discussed in Awbery (1984), Hannahs (2013), Iosad (2017a), and Bell (2018).

Table 35: Welsh mutations

C	Mutation Type		
	Soft	Nasal	Aspirate
p	b	mh	f
t	d	nh	θ
k	g	ŋh	χ
b	v	m	
d	ð	n	
g	Ø	ŋ	
t̪	l		
r̥	r		
m	v		

Finally, the stress system of Welsh deserves mention. Stress falls generally on the penult. There are a fair number of words with exceptional final stress and a very few items with antepenultimate stress. What's striking about the Welsh system is that English speakers routinely intuit stress as falling on the final syllable. There are two reasons for this. First, Welsh does not have stress-related vowel reduction so final stressless syllables are routinely not reduced. Second, as noted by Williams (1989) and Dogil & Williams (1999), the intonational system can, in some cases, give conflicting cues so that the final syllable surfaces with a fairly high pitch even when it isn't stressed.

5 General discussion

Table 36 provides a broad summary of interesting differences that emerge across the Celtic languages with respect to the phenomena discussed in this chapter.

The first row of Table 36 gives the number of mutations, showing that generally there are more of them in Brythonic than in Goidelic languages. The second row gives a crude characterization of the transparency of the mutations, where the mutations of the Goidelic languages can be seen as generally less transparent than those of the Brythonic languages. And, though it's not shown in Table 36, the Goidelic mutations also affect more consonants than those in the Brythonic languages. The next two rows show that the Goidelic languages generally exhibit vowel reduction and initial stress, whereas the Brythonic languages exhibit

Table 36: Summary of differences

	Goidelic			Brythonic		
	Irish	Manx	SG	Breton	Cornish	Welsh
Number of Mutations	2	1–2	2	4	4	3
Transparent Mutation	no	no	no	yes	yes	yes
V Reduction	yes	yes	yes	no	no	no
Stress Placement	initial*	initial	initial	penult	penult	penult
Grammatical Case	yes	?	yes	no	no	no
Palatalization Contrast	yes	yes	yes	no	no	no

penult stress with no or less vowel reduction. Finally, the last two rows show that overt grammatical case occurs in the Goidelic languages along with a palatalization contrast, while neither occur in the Brythonic languages.

Additionally, there are other commonalities described in this chapter that do not follow the Goidelic/Brythonic divide. For example, in all the languages, mutation is triggered by morphosyntactic environments. However, I also noted several cases of ‘phonological mutation’, namely the eclipsis mutation in Scottish Gaelic and the nasal mutation in Welsh. Finally, there is pre-stopping in virtually the same environment in two of the languages, Manx and Cornish.

Taking a closer look at the differences summarized in Table 36, one can observe that the phonology of the mutations is somewhat simpler in the Brythonic languages (as measured by transparency) than in the Goidelic languages – both in terms of the number of consonants affected and in terms of the mappings that occur. It would seem reasonable to suppose that this is balanced against the number of mutations in the Brythonic languages.

Another observation to ponder is that all the Goidelic languages exhibit a palatalization contrast whereas the Brythonic languages do not. One could ask if the absence of overt case in the Brythonic languages is connected to this, because case in the Goidelic languages is at least partially marked by palatalization.

Another potential point of covariance is vowel reduction and palatalization. The Goidelic languages exhibit both a palatalization contrast and vowel reduction whereas the Brythonic languages exhibit neither. Perhaps the way to look at these is a shifting of contrast from vowels to consonants in the Goidelic languages.

Of course, these observations are just hypotheses at this stage. To truly test them, these properties could be examined in the context of historical change and

language acquisition. For example, if these variables are connected as speculated above, I would expect them to arise diachronically in close succession. For example, is there evidence that vowel reduction arises when the stress systems separate? Similarly, I would expect to see these connections recapitulated in the timing of acquisition. For example, does the acquisition of the case system in the Goidelic languages correlate with the acquisition of the palatalization contrast?

In sum, these languages exhibit a range of interesting phonological properties. It is reasonable to see the variation across the systems as different ways of balancing phonological and morphological requirements and options. I've laid out several such possible connections in this chapter and propose that they can be tested with further study in the historical and language acquisition domains.

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